MATEUSZ POLITO

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Software Engineer with over 3 years of commercial experience in software development, test automation and embedded Linux. Focused on building reliable software that works with real hardware, from low-level protocols to internal tooling. Backed by a portfolio of projects in Rust, Python and web technologies.

SKILLS

Programming: Rust, Python, JavaScript/TypeScript, Bash *(advanced)*
C/C++, Java, SQL *(intermediate)*
Frameworks: tokio, Axum, React, Svelte, FastAPI, Robot Framework
Domains: Sensor simulation and fusion, binary protocols, 4G, 5G, (e)CPRI, I^2C
Systems & Infrastructure: Linux (including embedded), Docker, Git, nginx, Apache
AI tooling: Claude Code, GitHub Copilot, Cursor, Local LLMs
Languages: English (C2 – Cambridge), German (B1), Polish (Native)

WORK EXPERIENCE

Junior Rust Engineer @ Hertz New Technologies *Sep 2026 – present*

Developed and maintained two Rust codebases:

- **Simulator** for drones and drone detection devices:
  - A config-driven world with various types of drones and detection devices (Radar, RF, Optical)
  - Realistic device behaviour, based on ICDs and captured real data, down to the frame
  - Optional determinism (turning off random failures or misdetections) for reliable testing
  - Async tokio architecture: a world model which broadcasts its state over channels to independent tasks simulating various sensors, each with its own cadence and transport (TCP, RabbitMQ)
- **Fusion service** combining detections from multiple sensors into a single Kalman-filtered track per drone:
  - Channel-based tokio architecture with all state owned by one task, so no locks are needed
  - Associating point, area or angle-only detections from different sensors, while taking into account each sensor's stated accuracy
  - Capture and replay of real sensor traffic for reproducible testing and debugging

Integration Test Engineer @ Nokia *Aug 2022 – Mar 2024, Mar 2025 – Aug 2026*

- Developed and maintained automated Python and Robot Framework test suites validating CPRI/eCPRI communication and embedded Linux software updates for radio units.
- Built an internal Python tool monitoring RDP and SSH usage across test setups, preventing conflicts and unexpected session drops among integration team members.
- Maintained hardware test setups on-site (radio units, optical test devices, fibre cables), migrated them from Windows to Linux, and tested on simulated future hardware (SIMICS).

EDUCATION

M.Sc. Eng. in Applied Computer Science, Wrocław University of Science and Technology *Mar 2025 – Jul 2026*
Final grade: 5.0 / 5.0. Thesis: Automated benchmarking suite for ROS2 SLAM algorithms.

B.Eng. in Applied Computer Science, Wrocław University of Science and Technology *Oct 2021 – Jan 2025*

PROJECTS

- **Distributed Weather Station (Rust, ESP32, Raspberry Pi):** Monitoring node on ESP32, receiver program on Raspberry Pi, saving data to InfluxDB. Nodes communicated via shared message schema using serde.
- **Testing app for students (Svelte, TypeScript):** A web app for the Question DB format used at my uni, replacing legacy Java apps.
- **Home server (Docker, Rust, Android):** Self-hosted services, monitored by my Rust agent and phone/watch app.